United States Patent Application Publication 20250281119
Kind Code A1
Publication Date September 11, 2025
Inventor(s) ABE; Takahiro

MEDICAL SENSOR, AND SENSOR SUPPORTING MEMBER

Abstract

At least one light emitter is configured to emit light. At least one light detector is configured to output a signal corresponding to an amount of incident light. A supporting member has an adhesive face adapted to be attached to a body of a subject. A sheet is peelably attached to the adhesive face. The supporting member includes a first supporting portion, a second supporting portion and a third supporting portion. The first supporting portion supports and covers the light emitter. The second supporting portion supports and covers the light detector. The third supporting portion is located between the first supporting portion and the second supporting portion. The sheet is formed with a foldable portion extending across a portion facing the third supporting portion and including a portion whose cross-sectional shape is changed so as to increase foldability of the sheet.

Inventors: **ABE; Takahiro (Saitama, JP)**

Applicant: **NIHON KOHDEN CORPORATION** (Tokyo, JP)

Family ID: **96948205**

Assignee: **NIHON KOHDEN CORPORATION (Tokyo, JP)**

Appl. No.: **19/068531**

Filed: **March 03, 2025**

Foreign Application Priority Data

JP	2024-037121	Mar. 11, 2024

Publication Classification

Int. Cl.: **A61B5/00** (20060101); **A61B5/145** (20060101); **A61B90/00** (20160101)

U.S. Cl.:

CPC **A61B5/68335** (20170801); **A61B5/14542** (20130101); **A61B5/684** (20130101); **A61B90/08** (20160201); A61B2090/0807 (20160201); A61B2560/0462 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application is based on Japanese Patent Application No. 2024-037121 filed on Mar. 11, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

[0002] The present disclosure relates to a medical sensor provided with a light emitter that emits light, and a light detector that outputs a signal corresponding to an amount of incident light. The present disclosure also relates to a sensor supporting member for attaching the medical sensor to a body of a subject.

[0003] Japanese Patent Publication No. 2020-150996A discloses, as an example of a medical sensor, a sensor adapted to be attached to a fingertip of a subject to calculate a transcutaneous arterial oxygen saturation (SpO2) of the subject. The sensor includes a light emitter and a light detector. The light emitter emits light toward a tissue of the fingertip. The light having passed through the tissue is received by the light detector. The light detector outputs a signal corresponding to intensity of the detected light. The sensor includes a supporting member adapted to be wrapped around the fingertip. The supporting member supports the light emitter and the light detector. The signal that is outputted from the light detector is transmitted through a cable to a device that calculates SpO2.

SUMMARY

[0004] It is demanded to improve the convenience of a medical sensor configured such that a light emitter and a light detector are attached to a subject's body using a supporting member.

[0005] An illustrative aspect of the present disclosure may provide a medical sensor, comprising: [0006] at least one light emitter configured to emit light; [0007] at least one light detector configured to output a signal corresponding to an amount of incident light; [0008] a supporting member having an adhesive face adapted to be attached to a body of a subject; and [0009] a sheet peelably attached to the adhesive face, [0010] wherein the supporting member including: [0011] a first supporting portion supporting and covering the light emitter; [0012] a second supporting portion supporting and covering the light detector; and [0013] a third supporting portion located between the first supporting portion and the second supporting portion; and [0014] wherein the sheet is formed with a foldable portion extending across a portion facing the third supporting portion and including a portion whose cross-sectional shape is changed so as to increase foldability of the sheet.

[0015] An illustrative aspect of the present disclosure may provide a sensor supporting member adapted to attach, to a body of a subject, a medical sensor provided with at least one light emitter configured to emit light and at least one light detector configured to output a signal corresponding to an amount of incident light, the sensor supporting member comprising: [0016] a first supporting portion supporting and covering the light emitter; [0017] a second supporting portion supporting and covering the light detector; [0018] a third supporting portion located between the first supporting portion and the second supporting portion; [0019] an adhesive face adapted to be attached to the body; and [0020] a sheet peelably attached to the adhesive face, and formed with a foldable portion extending across a portion facing the third supporting portion and including a portion whose cross-sectional shape is changed so as to increase foldability of the sheet.

[0021] According to the configuration of each of the above illustrative aspects, as the part of the sheet is peeled from the supporting member, folding of the sheet is promoted at the foldable

portion, whereby not only the adhesive face can be caused to be partially exposed, but also it is possible to easily maintain a condition that the part of the sheet as peeled is separated from the adhesive face. As a result, not only attachment of the adhesive face as exposed to the body of the subject can be facilitated, but also it is possible to suppress occurrence of a situation that the adhesive face that is excessively exposed obstructs an initial stage of the attachment operation.
[0022] In addition, since a part of the sheet is not separated in accordance with the peeling operation, the part of the sheet as peeled may be used as a tab for peeling the remaining part of the sheet. Since it is not necessary to perform the operation of peeling the sheet from the edge of the supporting member multiple times, the operation of exposing the whole of the adhesive face can be smoothly performed. Furthermore, since the peeled sheet to be discarded is not separated into multiple pieces, the discarding operation can be smoothly performed as well.
[0023] Therefore, it is possible to improve the convenience of the medical sensor configured such that the light emitter and the light detector are attached to the body of the subject with the supporting member.

Description

DESCRIPTION OF THE DRAWINGS
[0024] FIG. **1** illustrates an appearance of a front face side of a sensor according to an exemplary embodiment.
[0025] FIG. **2** illustrates an appearance of a back face side of the sensor of FIG. **1**.
[0026] FIG. **3** illustrates an appearance of the sensor under a condition that a sheet of FIG. **2** is peeled off.
[0027] FIG. **4** illustrates how to attach the sensor of FIG. **1** to a finger of a subject.
[0028] FIG. **5** illustrates how to attach the sensor of FIG. **1** to the finger of the subject.
[0029] FIG. **6** illustrates how to attach the sensor of FIG. **1** to the finger of the subject.
[0030] FIG. **7** illustrates how to attach the sensor of FIG. **1** to the finger of the subject.
[0031] FIG. **8** illustrates how to attach the sensor of FIG. **1** to the finger of the subject.
[0032] FIG. **9** illustrates a cross-section of a foldable portion along a line IX of FIG. **2** as viewed from an arrowed direction.
[0033] FIG. **10** illustrates another exemplary cross-section of the foldable portion.
[0034] FIG. **11** illustrates another exemplary cross-section of the foldable portion.
[0035] FIG. **12** illustrates an appearance of a sensor according to another exemplary configuration.
DESCRIPTION OF EMBODIMENTS
[0036] Examples of embodiments will be described below in detail with reference to the accompanying drawings. In each of the drawings, the scale is appropriately changed in order to make each element as illustrated have a recognizable size.
[0037] FIG. **1** illustrates an appearance of a sensor **10** according to an exemplary embodiment. The sensor **10** is attached to a fingertip of the subject to calculate a blood light absorber concentration. The sensor **10** is an example of a medical sensor. The fingertip is an example of a body of a subject. Examples of the blood light absorber concentration include a percutaneous arterial oxygen saturation (SpO2).
[0038] The sensor **10** includes at least one light emitter **11**. The light emitter **11** is a semiconductor light emitter that emits light including a prescribed wavelength. Examples of the semiconductor light emitter include a light emitting diode and a laser diode. The number of light emitter **11** and the value of the wavelength are appropriately determined in accordance with the type of the blood light absorber concentration to be acquired.
[0039] The sensor **10** includes at least one light detector **12**. The light detector **12** outputs a detection signal corresponding to an amount of light detected on a light detecting face. Examples of

the light detector **12** include a photodiode, a phototransistor, and a photoresistor.

[0040] The sensor **10** includes a cable **13**. The sensor **10** is connected to a pulse photometer (not illustrated) through the cable **13**. The cable **13** accommodates a power supply line, a control signal line, and a detection signal line.

[0041] The power supply line and the control signal line are connected to a driving circuit (not illustrated) that is connected to the light emitter **11**. The power supply line supplies power from the pulse photometer to the drive circuit. The control signal line supplies a control signal transmitted from the pulse photometer to the driving circuit for controlling the activation or deactivation of the light emitter **11**.

[0042] The power supply line is also connected to the light detector **12**. Accordingly, the electric power from the pulse photometer is also supplied to the light detector **12** through the power supply line. The detection signal line electrically connects the light detector **12** and the pulse photometer. Accordingly, the detection signal outputted from the light detector **12** is transmitted to the pulse photometer, and subjected to appropriate processing.

[0043] The sensor **10** includes a supporting member **14**. FIG. **1** illustrates a front face side of the supporting member **14**. FIG. **2** illustrates a back face side of the supporting member **14**.

[0044] As illustrated in FIG. **1**, the supporting member **14** has a first supporting portion **141**, a second supporting portion **142**, and a third supporting portion **143**. The first supporting portion **141** supports and covers the light emitter **11**. The second support portion **142** supports and covers the light detector **12**. The third supporting portion **143** is located between the first supporting portion **141** and the second supporting portion **142**.

[0045] As illustrated in FIG. **2**, in the initial state, the back face side of the supporting member **14** is covered with a peelable sheet **15**. The sheet **15** has a foldable portion **151**. The foldable portion **151** extends across a portion of the supporting member **14** facing the third supporting portion **143**. As a result, the sheet **15** is sectioned into a first region **152** including a portion facing the first supporting portion **141** of the supporting member **14**, and a second region **153** including a portion facing the second supporting portion **142** of the supporting member **14**.

[0046] When a user peels the sheet **15** upon the usage of the sensor **10**, an adhesive face **144** illustrated in FIG. **3** is exposed. The adhesive face **144** has adhesiveness.

[0047] The adhesive face **144** includes a first translucent portion **144*a*** and a second translucent portion **144*b***. The first translucent portion **144*a*** is disposed so as to face the light emitter **11**. The second translucent portion **144*b*** is disposed so as to face the light detector **12**.

[0048] Each of the first translucent portion **144*a*** and the second translucent portion **144*b*** is formed of a material having transparency with respect to the wavelength of the light emitted from the light emitter **11**.

[0049] A method of attaching the sensor **10** to a finger **20** of a subject will be described with reference to FIGS. **1** to **8**.

[0050] As illustrated in FIG. **4**, a user peels the first region **152** of the sheet **15** away from the first supporting portion **141** of the supporting member **14**. As a result, a part of the adhesive face **144** located on a back face side of the first supporting portion **141** is exposed.

[0051] Subsequently, as illustrated in FIG. **5**, a nail side of the finger **20** of the user is placed so as to face the first supporting portion **141**. The adhesive face **144** located on the back face side of the first supporting portion **141** is adhered to the nail side of the finger **20**. Thereafter, the second region **153** of the sheet **15** is peeled off, so that a part of the adhesive face **144** located on back face sides of the second supporting portion **142** and the third supporting portion **143** is also exposed.

[0052] As illustrated in FIG. **1**, the supporting member **14** includes a first retaining portion **145**. The first retaining portion **145** extends leftward and rightward of the first supporting portion **141**. As illustrated in FIG. **3**, the adhesive face **144** includes a first adhesive face **144*c***. The first adhesive face **144*c*** is located on a back face side of the first retaining portion **145**.

[0053] As illustrated in FIG. **6**, the first retaining portion **145** is wrapped around the finger **20**. As a

result, the first adhesive face **144***c* is adhered to the finger **20**. **20**
[0054] Subsequently, the supporting member **14** is bent while the third supporting portion **143** is faced to a tip of the finger **20**, whereby the second supporting portion **142** is disposed so as to face a ball side of the finger **20**. The adhesive face **144** located on the back face side of the third supporting portion **143** is adhered to the tip of the finger **20**. The adhesive face **144** located on the back face side of the second supporting portion **142** is adhered to the ball side of the finger **20**.
[0055] As illustrated in FIG. **1**, the supporting member **14** includes a second retaining portion **146**. The second retaining portion **146** extends leftward and rightward of the second supporting portion **142**. As illustrated in FIG. **3**, the adhesive face **144** includes a second adhesive face **144***d*. The second adhesive face **144***d* is located on a back face side of the second retaining portion **146**.
[0056] As illustrated in FIGS. **7** and **8**, the second retaining portion **146** is wrapped around a portion of the supporting member **14** including the first retaining portion **145**. The second adhesive face **144***d* is adhered to the surface of the first retaining portion **145** while covering the surface. As a result, the retention of the sensor **10** to the finger **20** is finished.
[0057] In this state, the light emitter **11** and the light detector **12** are disposed so as to face each other with the finger **20** in between. The light emitted from the light emitter **11** that has passed through the first translucent portion **144***a* of the supporting member **14** is incident on the tissue of the finger **20**. The light that has passed through the tissue and the second translucent portion **144***b* of the supporting member **14** is incident on the light detecting face of the light detector **12**. The pulse oximeter performs processing for calculating a light absorber concentration in arterial blood contained in the tissue based on the detection signal outputted from the light detector **12**.
[0058] FIG. **9** illustrates a cross-section of the sheet **15** as viewed from the arrowed direction along the line IX illustrated in FIG. **2**. The foldable portion **151** includes a portion whose cross-sectional shape is changed so as to increase the foldability of the sheet **15**.
[0059] Specifically, the foldable portion **151** includes multiple grooves **151***a* at which the thickness of the sheet **15** is reduced. Each of the multiple grooves **151***a* is formed by, for example, scoring. As a result, the flexibility of the sheet **15** is enhanced while the first region **152** and the second region **153** are not separated from each other. The number of the groove **151***a* to be formed may be single.
[0060] According to such a configuration, as illustrated in FIG. **4**, as the first region **152** of the sheet **15** is peeled from the supporting member **14**, folding of the sheet **15** is promoted at the foldable portion **151**, whereby not only the adhesive face **144** located on the back face of the first supporting portion **141** can be caused to be partially exposed, but also it is possible to easily maintain a condition that the first region **152** as peeled is separated from the adhesive face **144**. As a result, not only attachment of the adhesive face **144** as exposed to the finger **20** can be facilitated, but also it is possible to suppress occurrence of a situation that the adhesive face **144** that is excessively exposed obstructs an initial stage of the attachment operation.
[0061] In addition, since a part of the sheet **15** is not separated in accordance with the peeling operation, the first region **152** as peeled may be used as a tab for peeling the second region **153**. Since it is not necessary to perform the operation of peeling the sheet **15** from the edge of the supporting member **14** multiple times, the operation of exposing the whole of the adhesive face **144** can be smoothly performed. Furthermore, since the peeled sheet **15** to be discarded is not separated into multiple pieces, the discarding operation can be smoothly performed as well.
[0062] Therefore, it is possible to improve the convenience of the sensor **10** configured such that the light emitter **11** and the light detector **12** are attached to the finger **20** with the supporting member **14**.
[0063] It should be noted that, as illustrated in FIG. **10**, the foldable portion **151** may be formed so as to include multiple through-holes **151***b*. In this case, the foldable portion **151** exhibits a perforated appearance. Even with such a configuration, the foldability of the sheet **15** can be enhanced while the first region **152** and the second region **153** are not separated from each other.

The number of through-hole **151**_b_ to be formed may be single.
[0064] Alternatively, as illustrated in FIG. **11**, the multiple grooves **151**_a_ may be formed without changing a substantial thickness of the sheet **15**. Even with such a configuration, the foldability of the sheet **15** can be enhanced while the first region **152** and the second region **153** are not separated from each other.
[0065] As illustrated in FIG. **1**, information for assisting attachment of the sensor **10** to the finger **20** is provided on the front face side of the supporting member **14**. Such information includes information indicating the position of the light emitter **11**, information indicating the position of the light detector **12**, information providing a guide indicating the position at which the finger **20** is to be placed, as well as information **147** providing a guide indicating the position at which the supporting member **14** is to be bent when the attachment to the finger **20** is performed (see also FIG. **6**). The foldable portion **151** of the sheet **15** is arranged so as to face the information **147** with the adhesive face **144** in between.
[0066] According to such a configuration, it is possible to promote association of the operation for folding the sheet **15** at the foldable portion **151** to partially expose the adhesive face **144** with the operation for attaching the supporting member **14** to the finger.
[0067] Each of the configurations exemplified above is merely illustrative for facilitating understanding of the present disclosure. Each exemplary configuration may be appropriately modified or combined with another exemplary configuration within the scope of the present disclosure.
[0068] As illustrated in FIG. **1**, the supporting member **14** according to the above exemplary embodiment has a shape that is axisymmetric with respect to a straight line L connecting the light emitter **11** and the light detector **12**. According to such a configuration, the attachment stability of the sensor with respect to the finger of the subject can be enhanced. However, as illustrated in FIG. **12**, the supporting member **14** may exhibit a belt shape
[0069] in which the light emitter **11** is supported at one end thereof. In this example, fixation of the sensor **10** is performed by wrapping the supporting member **14** around the finger of the subject. The light detector **12** is supported at a position facing the light emitter **11** with the finger in between when the supporting member **14** is wrapped around the finger.
[0070] The application of the medical sensor to which the concept of the present disclosure may be applied is not limited to the detection of the blood light absorber concentration. In the case of a medical sensor that optically detects a physiological parameter, the fundamental concept of the present disclosure can be applied through an appropriate shape change.

Claims

1. A medical sensor, comprising: at least one light emitter configured to emit light; at least one light detector configured to output a signal corresponding to an amount of incident light; a supporting member having an adhesive face adapted to be attached to a body of a subject; and a sheet peelably attached to the adhesive face, wherein the supporting member including: a first supporting portion supporting and covering the light emitter; a second supporting portion supporting and covering the light detector; and a third supporting portion located between the first supporting portion and the second supporting portion; and wherein the sheet is formed with a foldable portion extending across a portion facing the third supporting portion and including a portion whose cross-sectional shape is changed so as to increase foldability of the sheet.
2. The medical sensor according to claim 1, wherein the foldable portion includes a groove at which thickness of the sheet is reduced.
3. The medical sensor according to claim 1, wherein the foldable portion includes at least one through-hole.
4. The medical sensor according to claim 1, wherein a surface of the supporting member is

provided with information for assisting attachment to the body of the subject; and wherein the foldable portion is arranged so as to face the information with the adhesive face in between.

5. The medical sensor according to claim 1, wherein the supporting member exhibits an axisymmetric shape relative to a straight line connecting the light emitter and the light detector.

6. A sensor supporting member adapted to attach, to a body of a subject, a medical sensor provided with at least one light emitter configured to emit light and at least one light detector configured to output a signal corresponding to an amount of incident light, the sensor supporting member comprising: a first supporting portion supporting and covering the light emitter; a second supporting portion supporting and covering the light detector; a third supporting portion located between the first supporting portion and the second supporting portion; an adhesive face adapted to be attached to the body; and a sheet peelably attached to the adhesive face, and formed with a foldable portion extending across a portion facing the third supporting portion and including a portion whose cross-sectional shape is changed so as to increase foldability of the sheet.